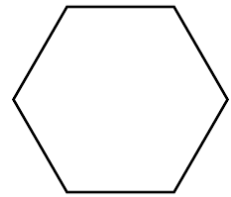


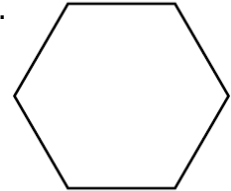
Grade 6 Accelerated
Unit 7
Lesson 7 Homework

Name _____
Date _____
Period _____

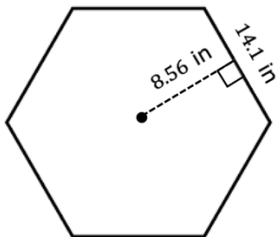
1. Decompose the regular hexagon into only triangles.



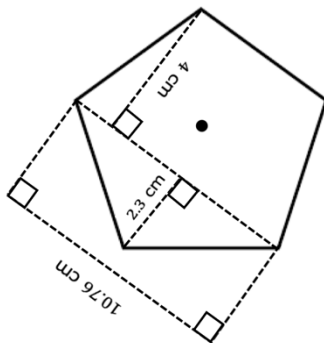
2. Decompose the regular hexagon into triangles and quadrilaterals.



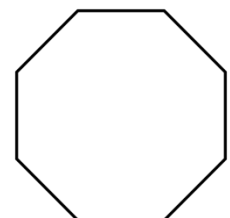
3. A hexagon is given in inches (in). Each side of the hexagon has a length of 14.1 inches and a perpendicular distance from the center to the midpoint of a side to be 8.56 inches. Find the area of the regular hexagon rounded to the nearest square inch.



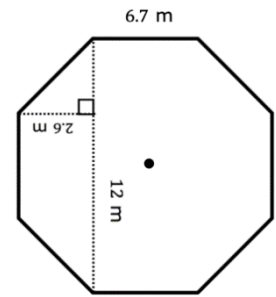
4. Find the area of the regular pentagon that has a side length of $5\frac{1}{2}$ cm.



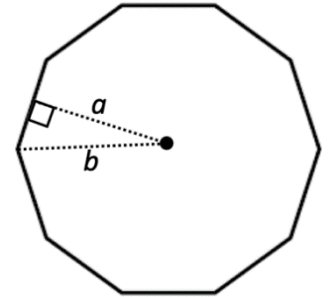
5. Decompose the regular octagon into three quadrilaterals.



6. An octagon is shown with lengths in meters (m). What is the area of the octagon?



7. Label the regular decagon with the following dimensions.
Side length = 6.2 in.
 $a = 7.4$ in.
 $b = 9.22$ in.

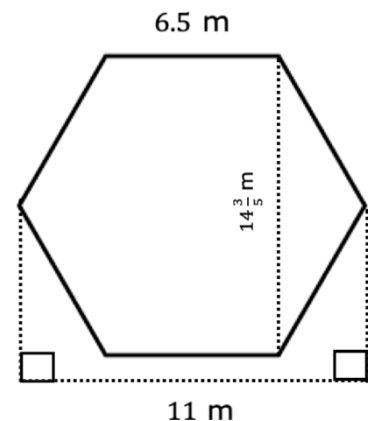


Which of the dimensions is not needed to find the area of the decagon?

8. Find the area of the decagon from Question 7 in square inches.

9. Myles found the area of the hexagon. His work is shown.

Myles' Work
Area of Trapezoid = $7.3(11 + 6.5)$ $A = 127.75 \text{ m}^2$ Area of hexagon = $2(127.75)$ $A = 255.5 \text{ m}^2$



Write a note to Myles explaining the error he made.

10. Find the area of the hexagon in square meters.